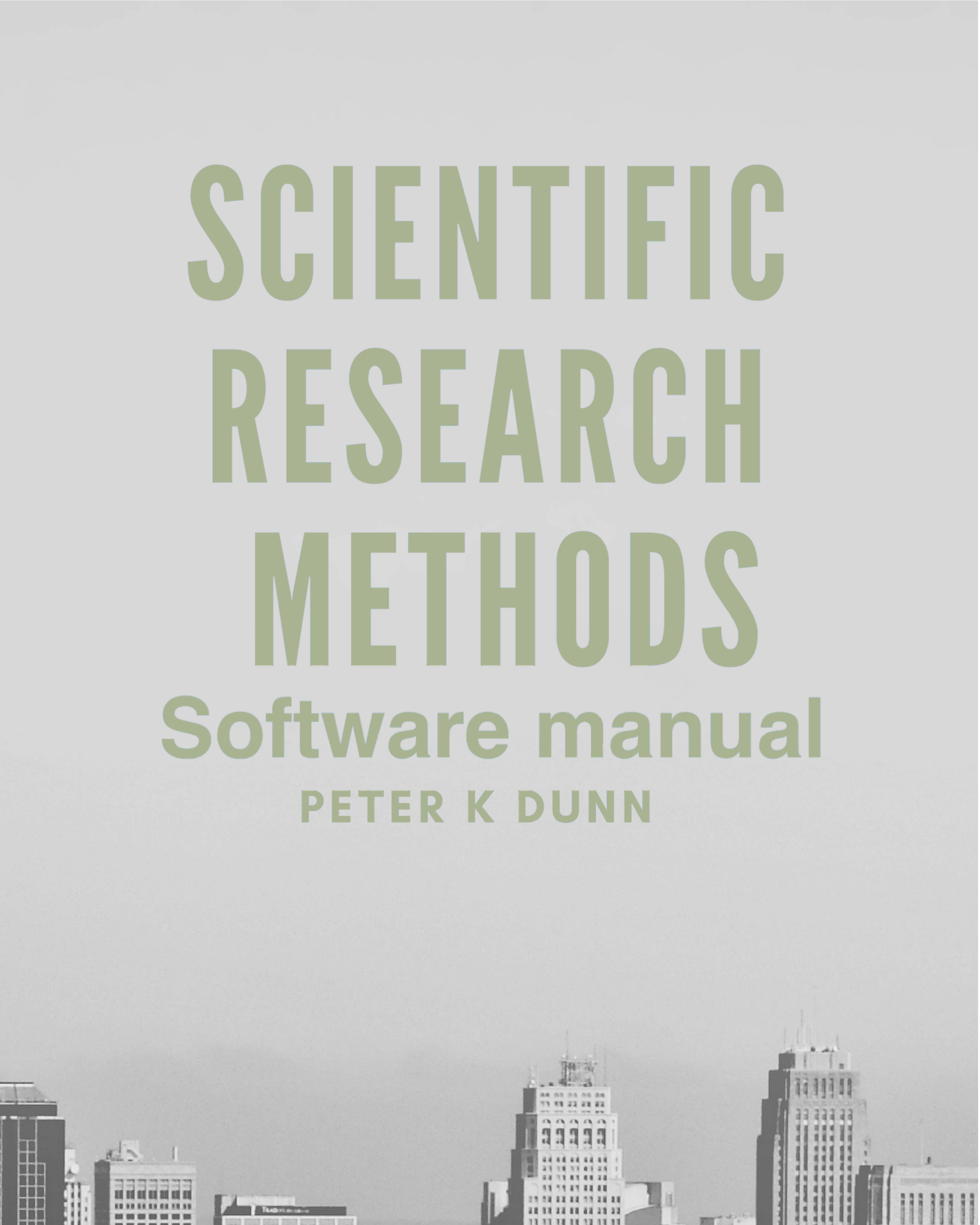

SCIENTIFIC
RESEARCH
METHODS
Software manual
PETER K DUNN

Peter K. Dunn

Science Research Methods: Software

An introduction to quantitative research and statistics: Using software

Contents

Preface

This book has been prepared for use with the book *Scientific Research and Methodology*, which is an introduction to quantitative research methods in the scientific, engineering and health disciplines.

This book introduces the whole research process, from asking a research question to analysis and reporting of the data. The focus, however, is on the analysis of data.

Statistical software

This manual explains how to conduct some of the analyses in the book *Scientific Research Method*[1] using software. We discuss using jamovi[2] (The jamovi Project, n.d.) and SPSS[3] (IBM Corp 2016) only.

jamovi is *free* and is like (but not exactly the same as) SPSS. From the jamovi homepage (sic)[4]:

> jamovi is a new '3rd generation' statistical spreadsheet. designed from the ground up to be easy to use, jamovi is a compelling alternative to costly statistical products such as SPSS and SAS.

Callouts used on this book

The following callouts are used in this book:

> These chunks introduce the objectives for the chapters of the book.

> These chunks offer helpful information.

> These chunks highlight common mistakes or warnings, about a particular concept or about using a formula.

> These chunks point to the data files being used, collated in Sect. 11, so you can download the data, and follow the instructions.

[1] https://peterkdunn.github.io/SRM-Textbook/
[2] https://www.jamovi.org/
[3] https://www.ibm.com/products/spss-statistics
[4] https://www.jamovi.org/

Using statistical software rather than a spreadsheet

(The following information is taken from Dunn (2025) (Sect. 1.5).)

Using spreadsheets for storing and analysing data requires care. Expensive and dangerous errors have been made due to using spreadsheets (AlTarawneh and Thorne 2017). Some challenges with using spreadsheets include that:

- spreadsheets may *change data* (e.g., reformatting entries as dates) when not appropriate (Ziemann et al. 2016).
- spreadsheets may include *formulas with errors* that are difficult to locate and hence fix (Panko 2016; London and Slagter 2021).
- spreadsheets *do not leave a record* of how the data were analysed or prepared. Keeping a record of the analysis, preparation of variables, and other operations with the data is good scientific practice (*reproducible research*) (Simons and Holmes 2019).
- spreadsheets often produce poor graphs (Su 2008).

Problems with spreadsheets, as with any software, are often due to human error, but *spreadsheets make errors hard to find and hence hard to fix*. Spreadsheets are useful for data collection and basic data manipulation, but are not designed for scientific analysis. Be careful using spreadsheets for research and analysis.

Using jamovi

We **strongly recommend** installing the most recent **stable**, **desktop** (not online) version of jamovi[5].

Using SPSS

All recent versions of SPSS work in similar ways. SPSS is installed in computer labs and available using *UniSC Anywhere*[6].

Learning Outcomes

In this book, you will learn to:

- Use jamovi to conduct chi-square tests.
- Use jamovi to conduct two-sample t-tests.
- Use jamovi to create graphs.
- Use SPSS to conduct chi-square tests.
- Use SPSS to conduct two-sample t-tests.
- Use SPSS to create graphs.

[5] https://www.jamovi.org/download.html
[6] https://Uniscanywhere.usc.edu.au

How to cite this book

Peter K. Dunn (2026). *Scientific Research Methods: Software manual*. https://peterkdunn.github.io/SRM-software/

1

Introduction

In this chapter, you will learn:

– introductory information about starting an analysis in software.

1.1 Before starting an analysis...

Before you start to use jamovi or SPSS, you need to know *why* you need the software. This document only discusses two scenarios:

- The **relationship between two qualitative variables**:
  - Analysis of **odds ratios** using, for example, a χ^2-test; or
  - Analysis of **two proportions** using, for example, a z-test.
  - Displays using a side-by-side or stacked barchart.
- The **difference between means in two groups**:
  - Analysis of two means using, for example, a two-sample t-test.
  - Displays using a error bar chart (and perhaps a boxplot).

The data files used in this document are available in Sect. 11. If you wish, you can download the data, and follow the instructions in this document.

1.2 The data: two qualitative variables

Consider data (Gammon et al. 2012) in Exercise 31.16 of the textbook[1] (shown here in Table 1.1).

The data file (`B12-Long.csv`) can be downloaded in Sect. 11. You can download the data, and follow the instructions.

The data has *two qualitative variables* being measured on each individual:

1. Whether the person is vegetarian (or not);
2. Whether the person is B12 deficient (or not).

Hence, jamovi will have *two columns*: one for each variable. There are 124 units of analysis, so there are 124 *rows* of data.

[1] https://peterkdunn.github.io/SRM-Textbook/AnalysisOddsRatio.html

TABLE 1.1: *The number of vegetarian and non-vegetarian women who are (and are not) B12 deficient.*

	B12 deficient	**Not B12 deficient**	**Total**
Vegetarians	8	26	**34**
Non-vegetarians	8	82	**90**
Total	**16**	**108**	**124**

1.3 The data: one qualitative, one quantitative variable

Consider the face-plant data (Wojcik et al. 1999) used in Exercise 30.11 of the textbook[2] (shown here in Table 1.2).

The data file (`ForwardFall.csv`) can be downloaded in Sect. 11.

You can download the data, and follow the instructions.

TABLE 1.2: *Lean-forward angles for older and younger women.*

Younger women		**Older women**
29	34	18
32	27	15
34	32	23
31	28	13
33	27	12

[2] https://bookdown.org/pkaldunn/SRM-Textbook/AnalysisTwoMeans.html

Part I

Using jamovi

2

jamovi: general information

In this chapter, you will learn:

– tips for working with jamovi.

2.1 About jamovi

- jamovi is a relatively **new** package (built on **R** (R Core Team 2025), which is very solid and reliable, and has been available since 1993).
- jamovi is **free** to download, from the jamovi webpage[1]; avoid the Cloud version.
- *Except for graphs*, jamovi output should **not** be presented in reports or presentations. Instead, the information from jamovi output should be used to construct proper tables (e.g., rounded numbers; units of measurement; decimal points aligned; etc.).
- The number of displayed decimal places (`dp`) or significant figures(`sf`) can be adjusted using the three vertical dots in the top right corner.

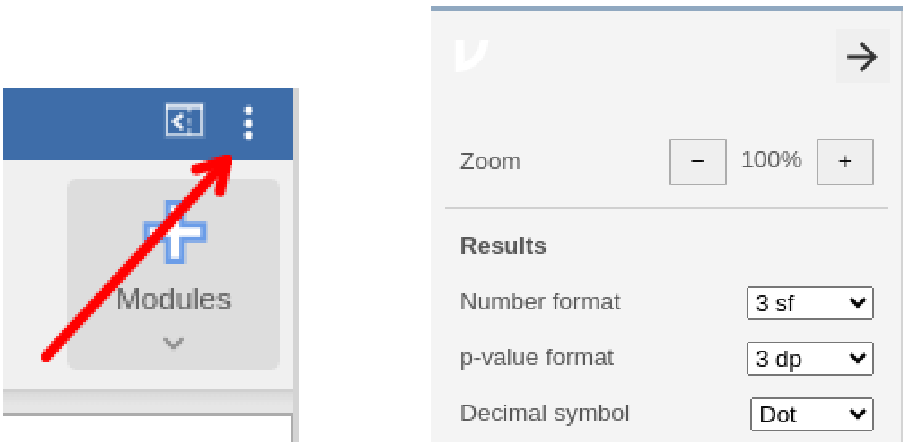


FIGURE 2.1: *Left: Click on the three dots to set the number of decimal places or significant figures shown. Right: The default settings*

2.2 Further help

You can get help with using jamovi by:

- Watching software screencasts on Canvas.

[1] https://www.jamovi.org/

- Using jamovi resources[2].
- Using LinkedIn Learning[3].
- Watching videos[4].
- Asking SCI110 staff **in tutorials**, or in the Canvas *Discussions*.

[2] https://www.jamovi.org/user-manual.html
[3] https://www.linkedin.com/learning/
[4] https://datalab.cc/tools/jamovi

3

jamovi: data set up and entry

In this chapter, you will learn:

- how to enter data correctly into jamovi.

3.1 Preparing for data entry

Before entering your data into jamovi, here are some **important reminders**:

- In jamovi (as with most statistical software), *each row represents one unit of analysis*.
- In jamovi (as with most statistical software), *each column represents one variable*.

3.2 Define the variable names

At the top of each column (that is: **above**, but *not in*, Row 1!) is the *name* of the variable. When you *double click* on the column headings (Fig. 3.1), information *about* the variables can be set (Fig. 3.2).

First, give the data a useful name. This name is what is used in plots and tables, for example.

Then you can tell jamovi information *about* the variables, such as whether the data are nominal, ordinal or 'continuous' (which is what jamovi calls *all* quantitative variables...), and the *levels* of the nominal and ordinal variables.

The information *about* the variables must be correctly set up **before** analysis.

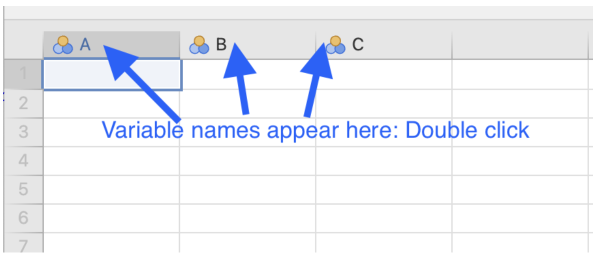


FIGURE 3.1: *Click on the column headers to describe the variables.*

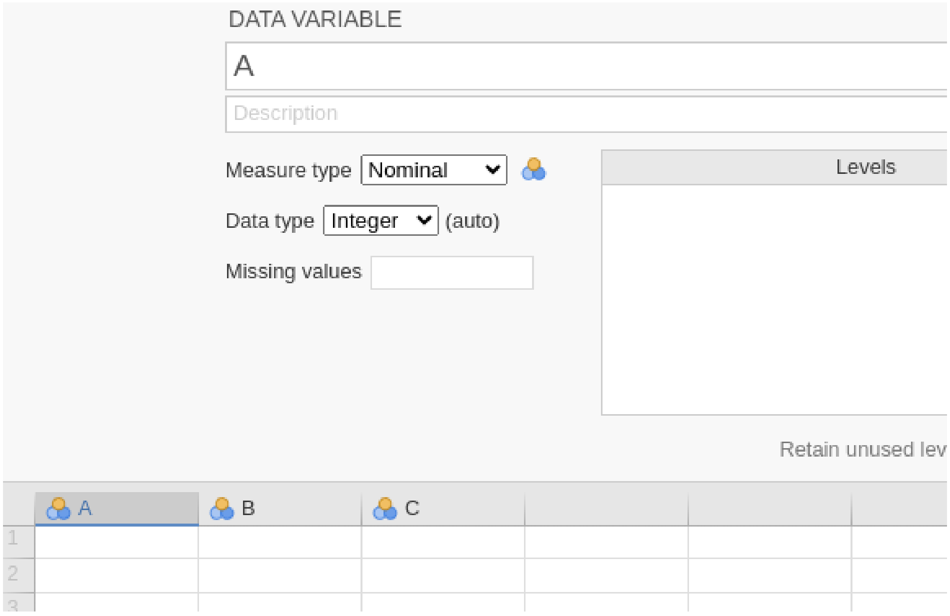


FIGURE 3.2: *The information to fill in for each variable.*

3.3 Define the variable information

For this section, the data introduced in Sect. 1.3 are used, as it demonstrates how to enter data for both *quantitative* and *qualitative* variables.

In the **Data** area, each row represents a unit of analysis (in this case, a person), as is usual in jamovi. For these data, each unit of analysis gets two variables recorded: one is quantitative (**Lean angle (in degrees)**), and one is qualitative (**Age group**). So there are two columns in the data, one for each variable (Fig. 3.3).

The order of the variables in the columns doesn't matter, but here the quantitative variable is listed first (the lean- angle), and then the qualitative variable (the age group).

The variables need to be properly set up by double-clicking on the variable names, as described below.

3.3.1 Adding information for quantitative variables

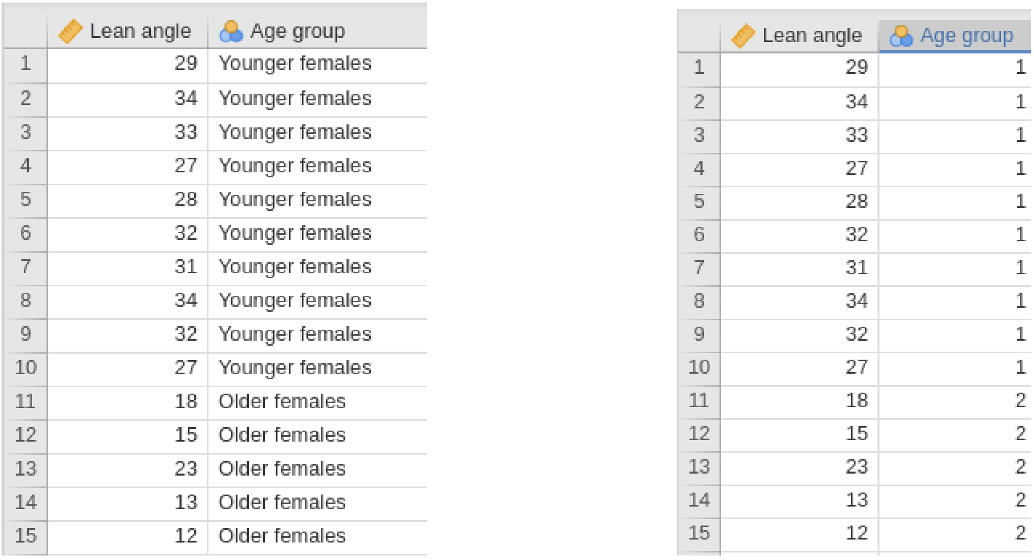
After double-clicking on the column header (the variable name) for a **quant**itative variable, set the variable type as **Continuous** (which, unfortunately, is how jamovi describes *all* quantitative data). The `Data type` can then be set as `Integer` (discrete) or `Decimal` (continuous). Also give the variable a sensible name (Fig. 3.4).

3.3.2 Adding information for qualitative variables

Qualitative data requires more care than for quantitative variable to set up, as the levels of the variable need to be described. After double-clicking on the column header (the variable name) for a **qual**itative variable, set the variable type as **Nominal** or **Ordinal** as appropriate.

The next step depends on *how* the data have been, or will be, entered into jamovi.

- If the data have been, or will be, entered by using the actual names of the levels (e.g., one level is labelled `Older females` and the other as `Younger females`) as in Fig. 3.3 (left image):



	Lean angle	Age group
1	29	Younger females
2	34	Younger females
3	33	Younger females
4	27	Younger females
5	28	Younger females
6	32	Younger females
7	31	Younger females
8	34	Younger females
9	32	Younger females
10	27	Younger females
11	18	Older females
12	15	Older females
13	23	Older females
14	13	Older females
15	12	Older females

	Lean angle	Age group
1	29	1
2	34	1
3	33	1
4	27	1
5	28	1
6	32	1
7	31	1
8	34	1
9	32	1
10	27	1
11	18	2
12	15	2
13	23	2
14	13	2
15	12	2

FIGURE 3.3: *The data entered in jamovi. Left: Using the actual names of the levels. Right: Using digits to represent the levels.*

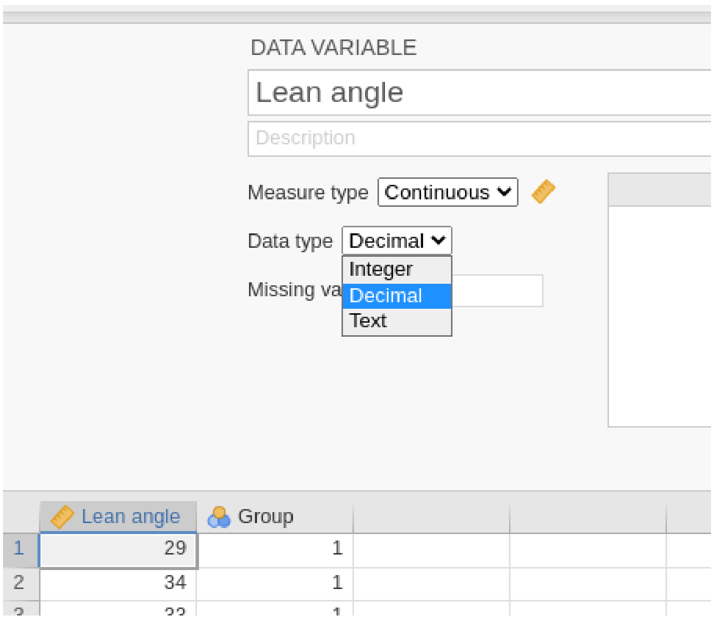


FIGURE 3.4: *Describing the quantitative variable.*

  - Set the `Data type` as `Text` (Fig. 3.5).
  - The data is then entered using these *exact* descriptions for the levels.
- If the data have been, or will be, entered by distinguishing the levels using digits (e.g., `1` is used for `Older females` and `2` is used for `Younger females`), as in Fig. 3.3 (right image)
  - Set the `Data type` as `Integer`, as in Fig. 3.6.
  - Then, jamovi must be told what these digits *mean*, by clicking on the individuals 'Levels' (Fig. 3.7), and replacing digits by what these levels mean. Notice, as shown in Fig. 3.6, that the integers are replaced by their *level meanings*, even though the digits remain in the data file (i.e., the data have not been changed).
  - The data is then entered using these *exact* digits for the levels.

The two age groups are defined so that a `1` represents younger women, and a `2` represents older women.

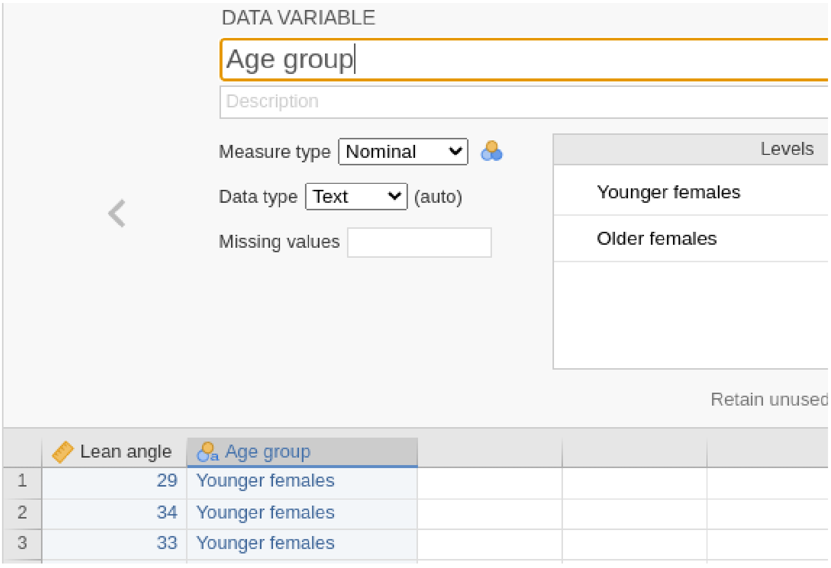


FIGURE 3.5: *Describing the qualitative variable. The first level has been described using text descriptions, such as 'Younger females'.*

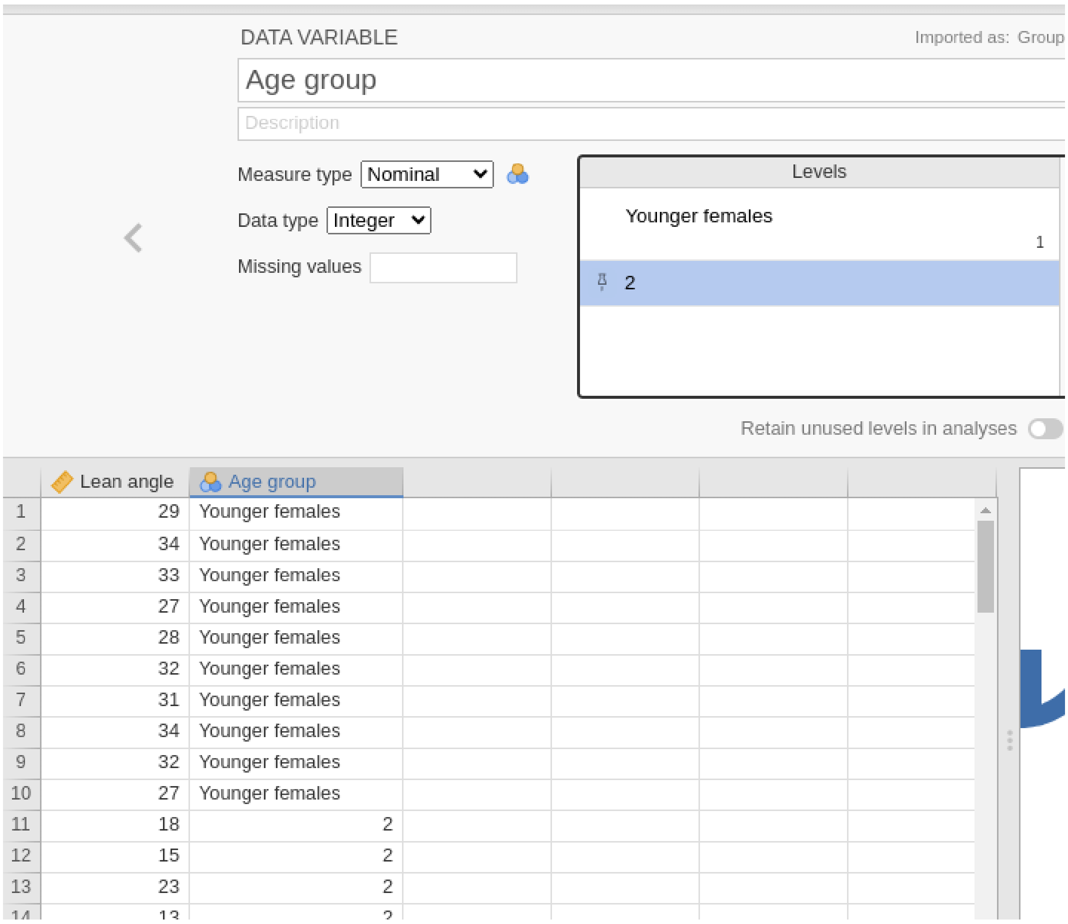


FIGURE 3.6: *Describing the qualitative variable: levels are distinguished using digits.*

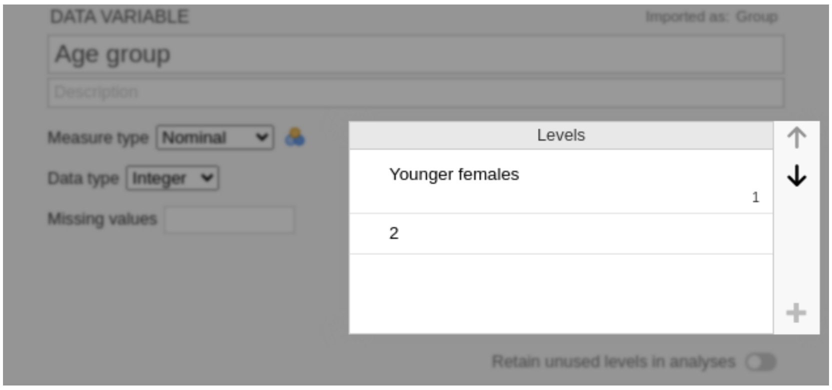


FIGURE 3.7: *Describing the qualitative variable.*

In the jamovi worksheet, the values `1` and `2` were entered, but jamovi shows what these mean in the worksheet when these values are defined.

3.3.3 Entering the data

Once the variables have been correctly defined, the data can be entered into the 'spreadsheet' area of the data file. For qualitative variable, make sure you enter the information correctly (as text, or as integers), as defined for that variable.

4

jamovi: creating and editing plots

In this chapter, you will learn to:

- Create plots in jamovi.
- Edit plots in jamovi.

The data files used in this section can be downloaded in Sect. 11.

You can download the data, and follow the instructions.

4.1 Introduction

Graphs in jamovi are **best created using the Plot menu** (though they can also be created as part of the **Analyses**, and sometimes through the **Exploration** menu).

The main advantage of using the **Plot** menu is that the plots can be properly edited.

4.2 Bar charts

4.2.1 Creating bar charts

For this section, the data introduced in Sect. 1.2 are used.

1. **Use the jamvoi menu:** select **Plot** and select **Bar Plot** (Fig. 4.1).
2. **Add the variables**. Add the variables to the rows and columns (in this case, the `Categorical Variable` is `Diet`, and the `Grouping Variable` is `B12`), as in Fig. 4.1.

Your graph will appear in the jamovi Output window.

3. To change the **type** of bar chart (i.e., stacked barchart or side-by-side barchart), select the `Group Options` and then the type of bar chart you would like, as in Fig. 4.2:

Making general changes—such as editing axis labels and titles—are discussed in Sect. 4.5.

4.3 Boxplots

4.3.1 Creating boxplots

For this section, the data introduced in Sect. 1.3 are used.

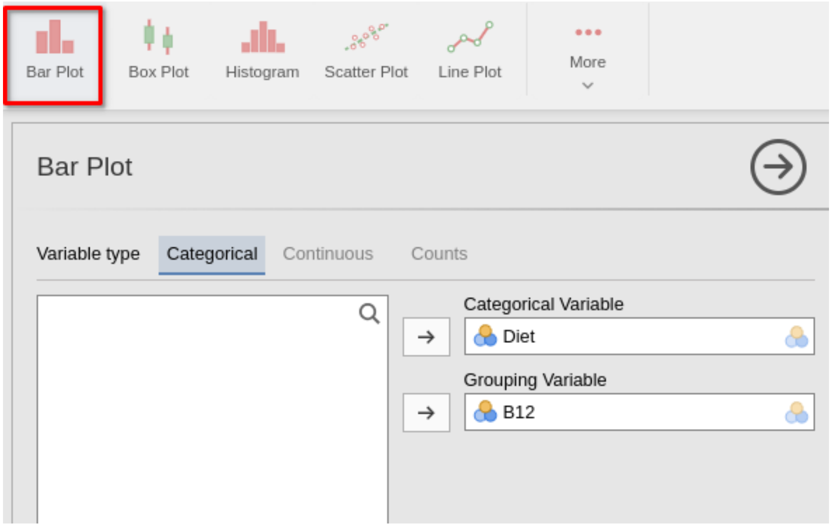


FIGURE 4.1: *Adding the variables to create a bar chart in jamovi.*

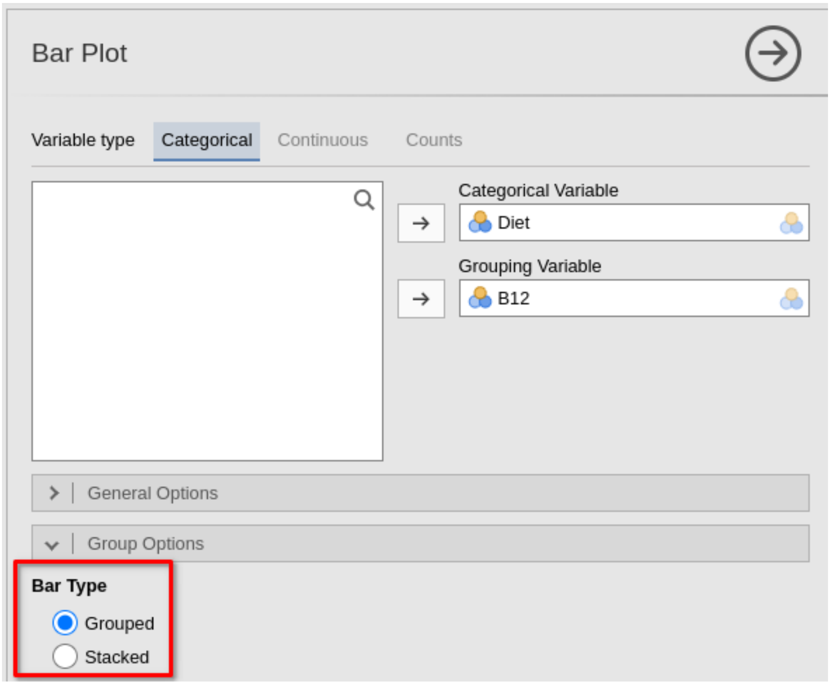


FIGURE 4.2: *Selecting the type of bar chart.*

1. **Use the jamovi menu:** select **Plot** and select **Box Plot** (Fig. 4.3).
2. **Add the variables**. Place the quantitative variable in the *Variables* area, and the qualitative variable in the **Grouping variable 1** area (Fig. 4.4).

Your graph will appear in the jamovi Output window.

4.3.2 Editing boxplots

Some boxplot-specific options can be edited by selecting the **General Options**. You probably will not need to change any of these options.

Making general changes—such as editing axis labels and titles—are discussed in Sect. 4.5.

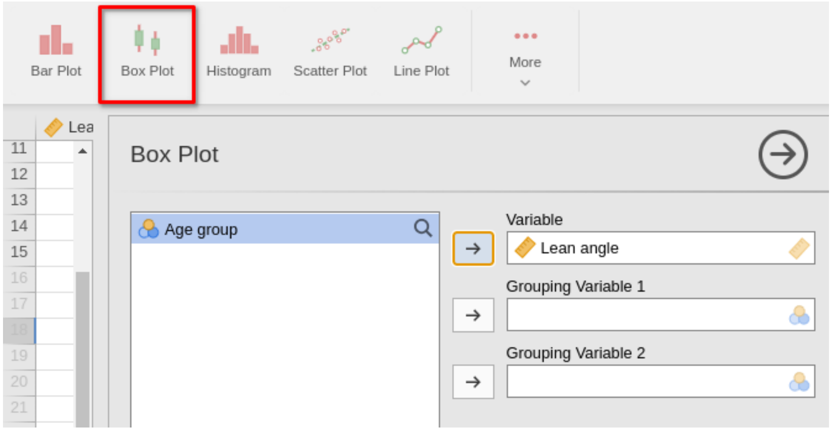


FIGURE 4.3: *Selecting a boxplot.*

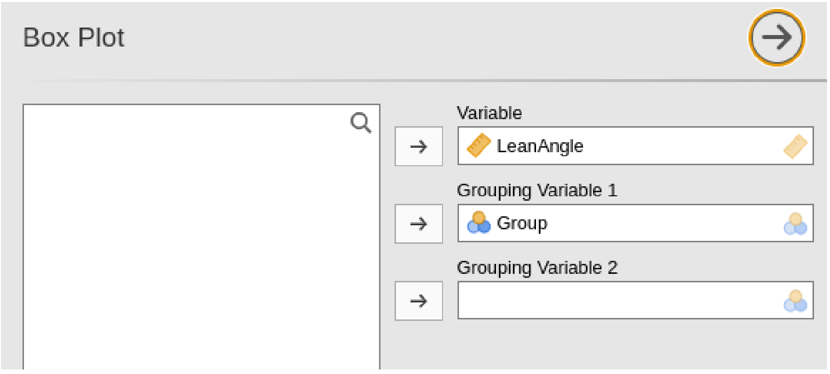


FIGURE 4.4: *Selecting the variables for a boxplot.*

4.4 Error bar charts

Error bar charts are produced as part of the *analysis*: see Chap. 6. These cannot be edited further.

4.5 Editing plots

The options to **edit** bar charts and boxplots are at the bottom of the plot window (Fig. 4.5).

Clicking on the options reveals the things that can be changed (Fig. 4.6):

- **Plot and Axis Titles**:
  - The **Plot Title**: The plot title must be *meaningful* and *contextual*
  - The **Plot Subtitle**: The plot subtitle may not be necessary. If you do add a subtitle, it must be *meaningful* and *contextual*
  - The **Plot Caption**: The *caption* can be added here, or can be added in Word or PowerPoint etc. after pasting the plot into your document.
  - The **X-Axis Title**: The label on the horizontal (`X`) axis must be *meaningful* and *contextual*.
  - The **Y-Axis Title**: The label on the vertical (`Y`) axis must be *meaningful* and *contextual* (and including units of measurement when appropriate).
- **Axes**: You *may* also need to adjust the upper and lower limits of the `Y` axis (called the **Range**)

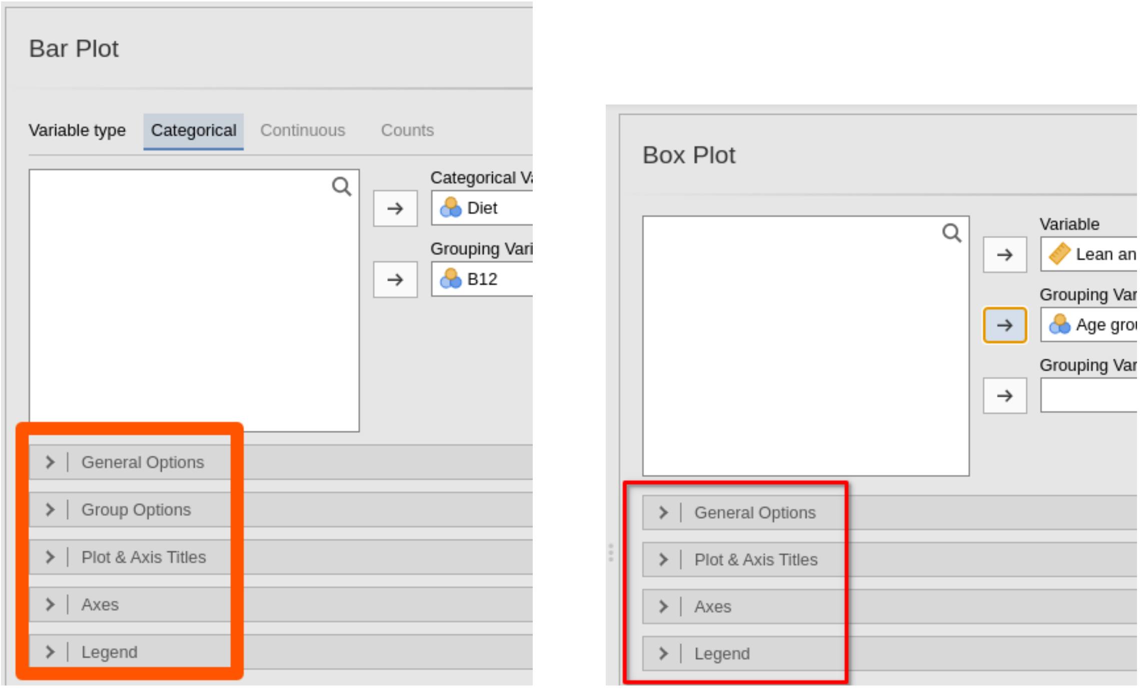


FIGURE 4.5: *The plotting options for bar charts (left) and boxplots (right).*

- **Legend**: The plot legend can be adjusted here if necessary. Importantly, the legend title must be *meaningful* and *contextual*.

Your changes will be evident in plot in the jamovi Output window.

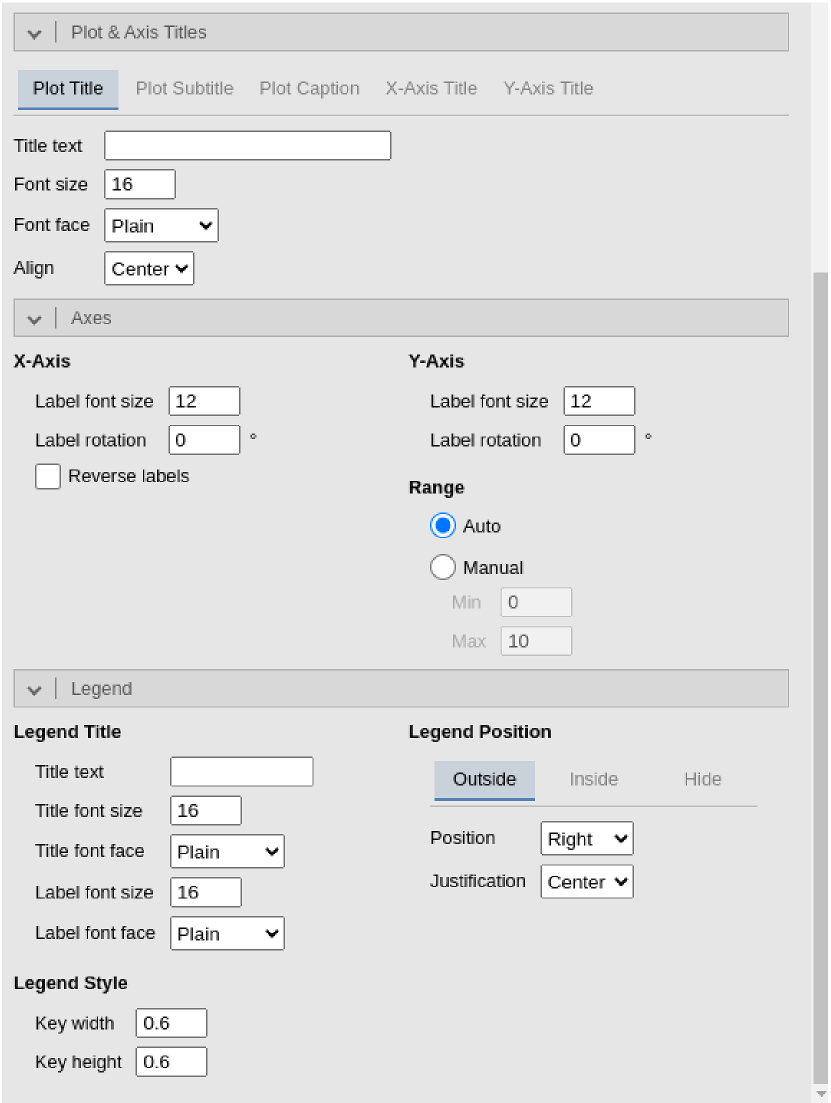


FIGURE 4.6: *The plotting options expanded for bar charts*

5

jamovi: inference for two odds or two proportions

In this chapter, you will learn to:

- use jamovi for comparing two odds (i.e., analysis using odds ratios).
- use jamovi for comparing two proportions.

5.1 Data entry

For this section, the data introduced in Sect. 1.2 are used. The data should be set up as described in Chap. 3. In the data:

- **Diet**: `1` means 'Vegetarian' (i.e. in Row 1 of the data table), and `2` means 'Non-vegetarian' (i.e. in Row 2 of the data table in Table 1.1).
- **B12**: `1` means 'B12 deficient' (i.e. in Column 1 of the data table), and `2` means '*Not* B12 deficient' (i.e. in Column 2 of the data table in Table 1.1).

The data needs to be set up so that *each row represents one unit of analysis*. That means there are 124 rows of data: one for each person (unit of analysis).

5.2 Analysis

This data set is available in Appendix 11. Once the data are entered or loaded, we can begin analysis:

1. **Use the menu:** select **Analyses> Frequencies> Independent samples (χ^2 test of association)** (Fig. 5.1).
2. **Select the variables:** dragging them from the left panel to the **Rows** and **Columns** areas as appropriate (Fig. 5.2). In Table 1.1, **Diet** is in the rows of the table, for example.
3. **Select the analysis options:** Under the area where you entered the variables, select the right-pointing arrow beside **Statistics**, and then (Fig. 5.3):
   - Select `X2`: this conducts the χ^2 hypothesis test for comparing odds.
   - Select `z test for difference in 2 proportions`: this conducts the z-test to compare proportions.
   - Select `Odds ratio`: this shows the value of the odds ratio.
   - Select `Difference in proportions`: this shows the difference between the two proportions.
   - Select `Confidence intervals`: this shows a CI for the odds ratio, and a CI for the difference between the proportions.

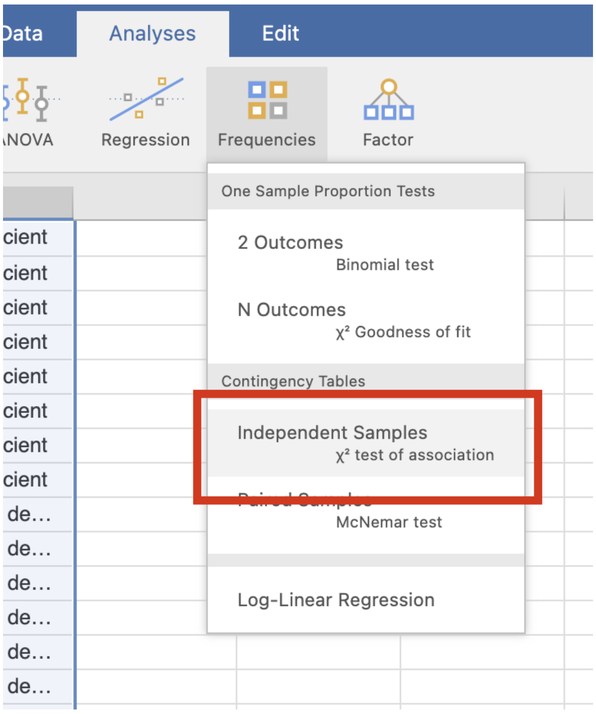


FIGURE 5.1: *Select the correct test.*

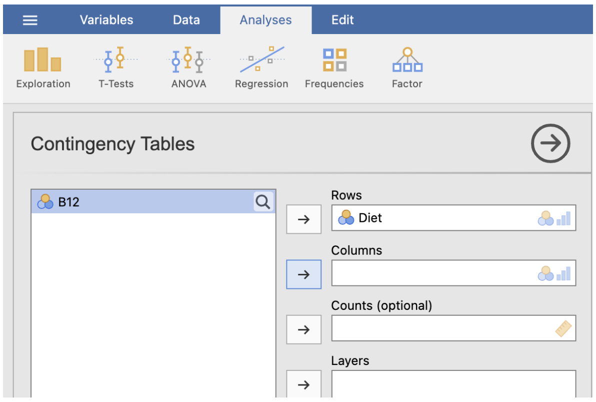


FIGURE 5.2: *Enter the variables.*

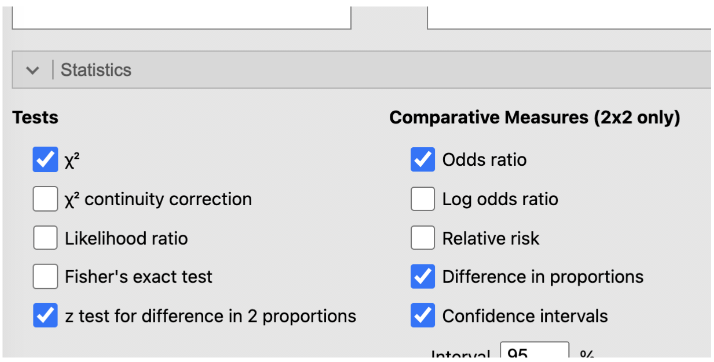


FIGURE 5.3: *Check the correct options.*

4. You will have your output in the jamovi Output window on the right.

5.3 Numerical summaries

Percentages can be obtained from the same place: **Analyses> Frequencies Independent samples** (χ^2 **test of association)**. Under the area where you selected the **Statistics**, select the right-pointing arrow beside **Cells** (Fig. 5.4), and then:

- Select `Row` under the **Percentages** section; and/or
- Select `Column` under the **Percentages** section.

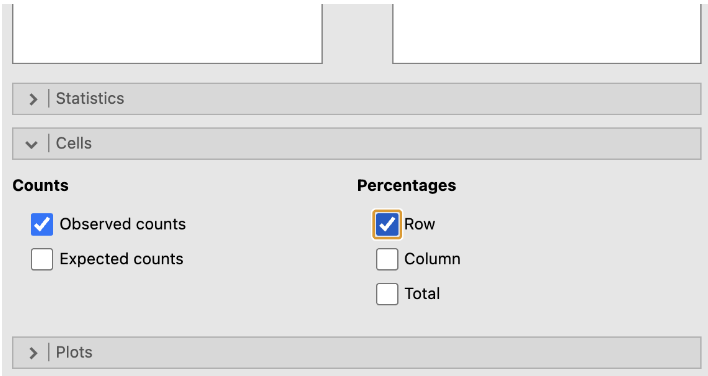


FIGURE 5.4: *Select what to show in the table.*

Of course, you can select any of the other options too, and see what happens.

6

jamovi: inference about the difference between two means

In this chapter, you will learn to:

- use jamovi for comparing two means.

6.1 Data entry

For this section, the data introduced in Sect. 1.3 are used. In the **Data** area, each row represents a unit of analysis (in this case, a person), as is usual in jamovi. The data should be set-up as in Sect. 3. In the data:

- **Lean angle**: the lean-forward angle for each woman (**quant**itative).
- **Age group**: the age group for each woman (**qual**itative).

The data needs to be set up so that each row represents one unit of analysis. That means there are 15 rows of data: one for each person (unit of analysis).

6.2 Analysis

This data set is available in Appendix 11. Once the data are entered or loaded, we can begin analysis:

1. **Use the jamovi menu**: select **Analyses> T-Tests> Independent Samples T-Test** (Fig. 6.1).
2. **Select the Variables**.
   - For the **Dependent Variables**, add the quantitative variable (here, `LeanAngle`).
   - For the **Grouping Variable**, add the qualitative variable (here, `Group`).
3. Then select the options:

- `Student's` and `Welch's` should be select to perform the hypothesis test and compute the CIs.
- `Mean difference` to show the difference between the two means in the output.
- `Confidence interval` to show the CI for the difference between the two means.
- `Descriptives` to show the numerical summary of each group.
- `Descriptives plot` to show the error bar chart.

Even if you have a one-tailed hypothesis, keep the hypotheses as the two-tailed (first) option, or you will get some strange output. The one-tailed P-value will be half the two-tailed P-value.

4. You will have your output in the jamovi Output window.

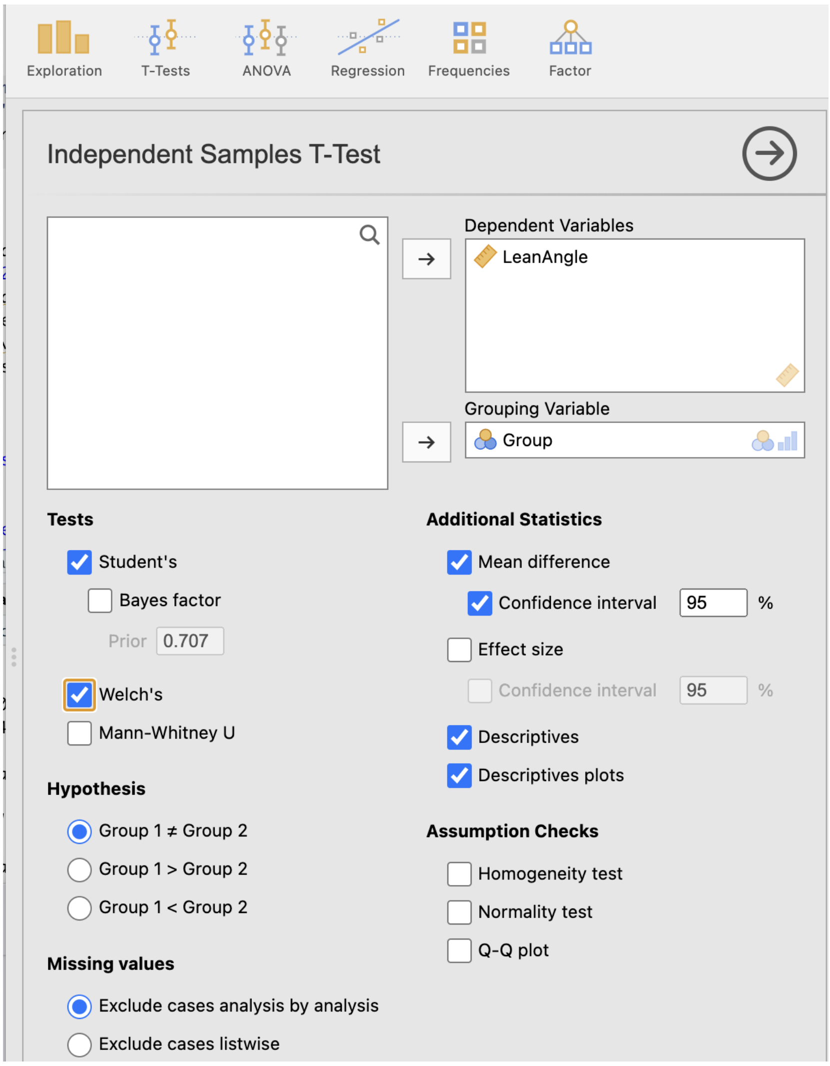


FIGURE 6.1: *Enter the variables.*

6.3 Numerical summaries

The *mean*, *median*, *standard deviation* and *standard error* of each group are produced as part of the output from the t-test analysis step above.

The *mean* and *standard error* for the *difference* is obtained from the t-test output too.

Part II

Using SPSS

7

SPSS: general information

In this chapter, you will learn:

- Tips for working with SPSS.

Before starting analysis with SPSS (IBM Corp 2016), here are some **general tips**:

- In general, you cannot download a copy of SPSS for use at home (which you *can* do with jamovi).
- *Except for graphs*, SPSS output should **not** be presented in reports or presentations. Instead, the information from SPSS output should be used to construct proper tables, or to produce proper conclusions.
- SPSS has two worksheets sheets; **both** must be set up correctly (you can switch between the two worksheets by clicking on these at the bottom of the SPSS window):
  - The **Data View**, which contains the data.
  - The **Variable View**, which contains information *about* each variable.
- In SPSS (as with most statistical software), *each row represents one unit of analysis* in the **Data View**.
- In SPSS (as with most statistical software), *each column represents one variable* in the **Data View**.
- At the top of each column (but **not** in Row 1!) is the *name* of the variable;
- When setting up the **Variable View**, many columns can be filled in (Fig. 7.1). These are the important columns to get right (the others aren't important for our purposes):
  - **Name**: A short description of the variable (with no spaces or punctuation!); for example `DiaBP`.
  - **Type**: Usually **Numeric** (apart from *names* which can be **String**).
  - **Label**: A fuller description of the variable. This is what will appear in tables and graphs to describe the variable. For example `Diastolic blood pressure (in mm Hg)`.
  - **Values**: For *qualitative* variables only: tell SPSS what each number represents (for example, does a `1` represent females, or males?)
  - **Measure**: *This is important*: You must tell SPSS whether each variable is nominal, ordinal, or scale (i.e. quantitative) by using the drop-down options.
- You can get help with using SPSS by:
  - Using LinkedIn Learning[1].
  - Watching software screencasts on Canvas.
  - Borrowing books in the library.
  - Asking SCI110 staff, including using the Canvas *Discussions*.

Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role

***FIGURE 7.1:** Information that can be given for each variable in the Variable View window).*

[1] https://www.linkedin.com/learning/

8

SPSS: inference for two odds or two proportions

In this chapter, you will learn to:

- use SPSS for comparing two odds (i.e., analysis using odds ratios).
- use SPSS for comparing two proportions.

8.1 Introduction

χ^2 tests ('*ki*-squared tests') are used to analyse data where each unit of analysis has *two qualitative variables* measured (or observed) about it. This type of data can be compiled into a two-way table of counts.

Consider data (Gammon et al. 2012) used in Exercise 31.16 of the textbook[1] (Table 8.1).

Two qualitative variables being measured on each individual:

1. Whether the person is vegetarian (or not);
2. Whether the person is B12 deficient (or not);

So SPSS will have *two columns*: one for each variable.

The data file (`B12-Long.sav`) can be downloaded in Sect. 11.

You can download the data, and follow the instructions.

TABLE 8.1: *The number of vegetarian and non-vegetarian women who are (and are not) B12 deficient.*

	B12 deficient	**Not B12 deficient**	**Total**
Vegetarians	8	26	**34**
Non-vegetarians	8	82	**90**
Total	**16**	**108**	**124**

8.2 Data entry

In the SPSS **Variable View**:

- **Diet**: `1` means 'Vegetarian' (i.e. in Row 1 of the data table), and `2` means 'Non-vegetarian' (i.e. in Row 2 of the data table).

[1] https://peterkdunn.github.io/SRM-Textbook/AnalysisOddsRatio.html

- **B12**: 1 means 'B12 deficient' (i.e. in Column 1 of the data table), and 2 means '*Not* B12 deficient' (i.e. in Column 2 of the data table).

See Fig. 7.1.

The usual approach for data entry used by SPSS is that *each row represents one unit of analysis*. That means that there will be 124 rows of data: one for each person (unit of analysis).

In SPSS, the data are entered as shown in Fig. 8.1, but there would be 124 rows.

Diet	B12
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	2
1	2
1	2
1	2

FIGURE 8.1: *Data entered into SPSS.*

The **Variable View** would describe these two variables; see Fig. 8.2.

	Name	Type	Width	Deci...	Label	Values	Mis...	C...	Align	Measure	Role
1	Diet	Numeric	8	0	Diet	{1, Vegetarian}...	None	8	Ri...	Nominal	Input
2	B12	Numeric	8	0	B12 Deficiency	{1, B12 deficient}...	None	8	Ri...	Nominal	Input

FIGURE 8.2: *Describing the variable in the Variable View window.*

8.3 Analysis

Once the data are entered, we can begin analysis:

1. **Use the menu:** select **Analyze> Descriptive Statistics> Crosstabs...**
2. **Select the variables**, dragging them from the left panel to the **Row(s)** and **Column(s)** areas as appropriate. In Table 8.1, **Diet** is in the rows of the table, for example. See Fig. 8.3.
3. **Select the analysis options.** Click the **Statistics...** button, and then (Fig. 8.4):
   - Select the **Chi-square**: to produce information for the hypothesis test.
   - Select the **Risk** option: to produce a CI for the odds ratio.
4. Press **Continue**, and then press **OK**. You will have your output in the SPSS Output window.

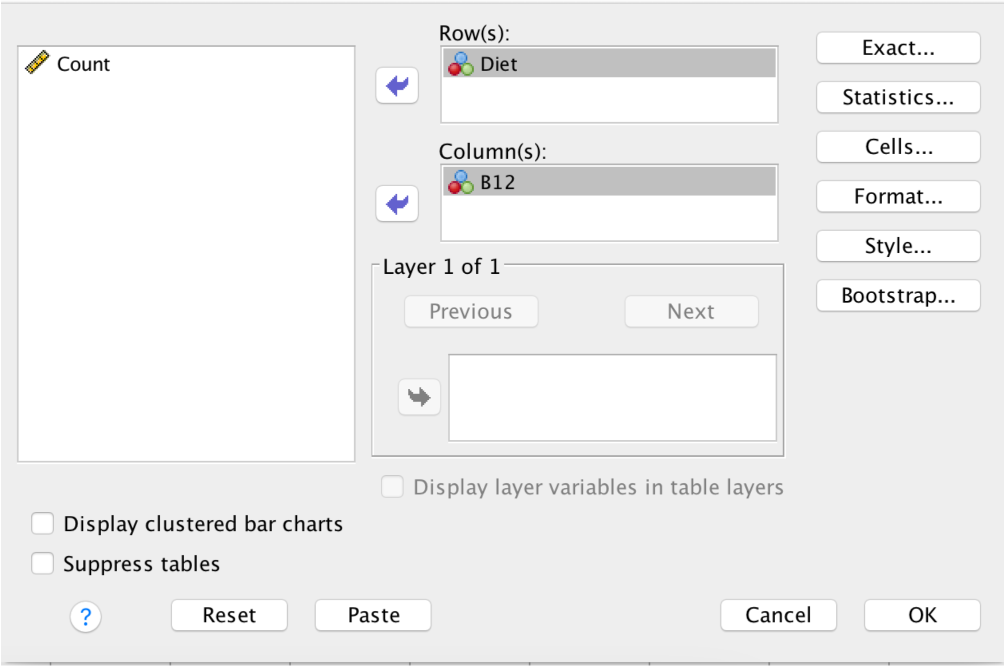


FIGURE 8.3: *Add the variables.*

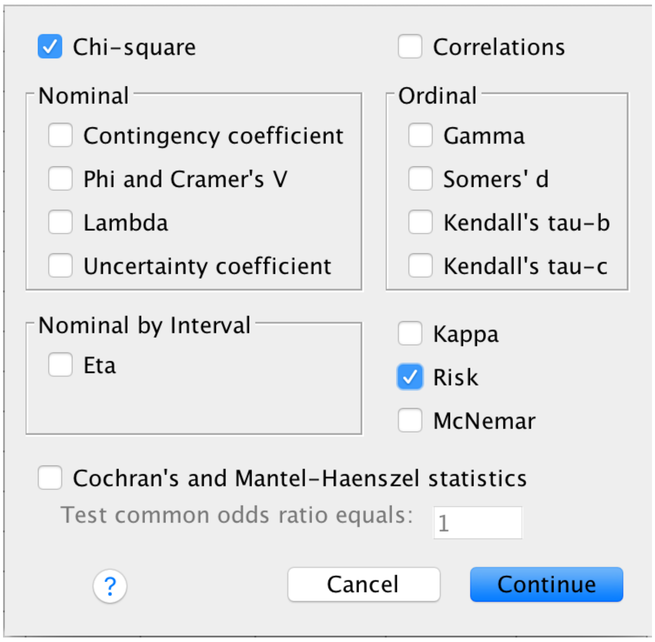


FIGURE 8.4: *Select the options.*

8.4 Numerical summaries

Percentages can be obtained from **Analyze> Descriptive Statistics> Crosstabs…** In the window that appears, select **Cells…** and then select **Percentages** as **Row**, as **Column**, or even as both if you wish.

Odds are computed as part of the inference output from Sect. 8.3.

9

SPSS: inference about the difference between two means

In this chapter, you will learn to:

- use SPSS for comparing two means.

9.1 Introduction

Two independent sample t-tests are used to compare the means of two separate groups.

Consider the face-plant data (Wojcik et al. 1999) used in Exercise 30.11 of the textbook[1] (Table 9.1).

TABLE 9.1: *Lean-forward angles for older and younger women.*

Younger women		Older women
29	34	18
32	27	15
34	32	23
31	28	13
33	27	12

The data file (`ForwardFall.csv`) can be downloaded in Sect. 11.

You can download the data, and follow the instructions.

9.2 Data entry

In the **Data View**, each row represents a unit of analysis (in this case, a person), as is usual in SPSS.

Each unit of analysis gets two variables recorded on it: one is quantitative (`LeanAngle`), and one is qualitative (`Group`). So there are two columns in the **Data View**, one for each variable.

The order doesn't matter, but here the quantitative variable is listed first (the lean-forward angle), and then the qualitative variable (the age group).

The **Variable View** needs to be set up correctly too (Fig. 9.1). For example, all variables should be given a descriptive label. The quantitative variable (here, `LeanAngle`) should be declared as **Scale** in the **Measure** column. The qualitative variable (here, `Group`) should be declared as **Nominal** in the **Measure** column (Fig. 9.2).

[1] https://peterkdunn.github.io/SRM-Textbook/AnalysisTwoMeans.html

LeanAngle	Group
29	1
34	1
33	1
27	1
28	1
32	1
31	1
34	1
32	1
27	1
18	2
15	2
23	2
13	2
12	2

FIGURE 9.1: *The data in the Data View.*

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure
1	LeanAngle	Numeric	8	0	Maximum lean...	None	None	11	Right	Scale
2	Group	Numeric	8	0		{1, Younger ...	None	8	Right	Nominal

FIGURE 9.2: *The Variable View.*

The two age groups are defined so that a `1` represents younger women, and a `2` represents older women. Since this is arbitrary, we tell SPSS this by setting the **Values** in the **Variable View** (Fig. 9.3).

9.3 Analysis

1. **Use the SPSS menu**: select **Analyze> Compare Means> Independent Samples T Test...**
2. **Select the Variables**. The **Test Variable** is the quantitative variable (here, `LeanAngle`), and the **Grouping Variable** is the qualitative variable (here, ‘`Group`); see Fig. 9.4.
3. **Define the groups to compare**. Click the button **Define Groups...** and enter the numbers that are being used to represent the two groups to be compared. In this example, the two groups are given the values `1` and `2` in the **Data View**, so we enter these.
4. Click the **Continue** button, then **OK**. You will have your output in the SPSS Output window.

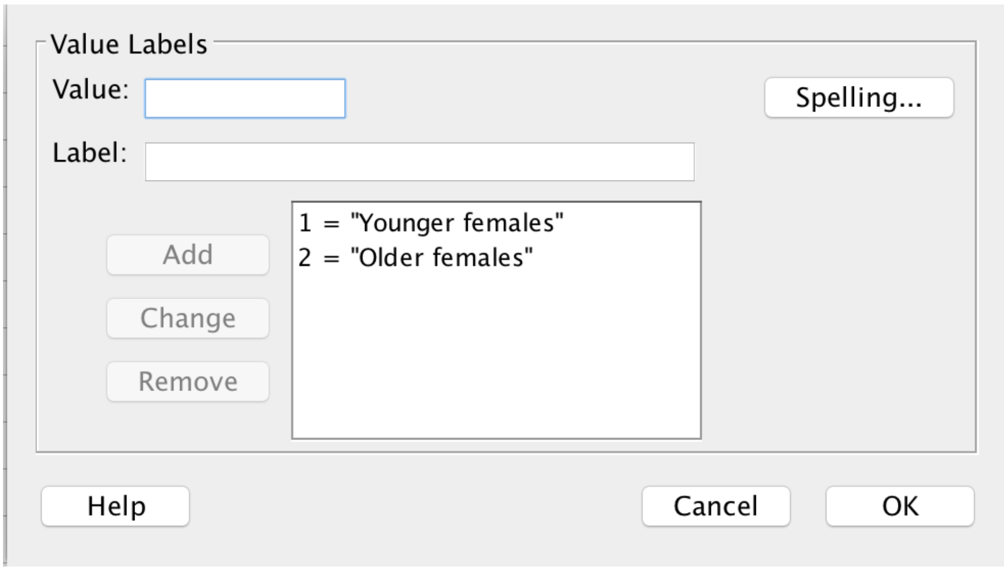


FIGURE 9.3: *Defining the levels (Values) for the age group, in the Variable View.*

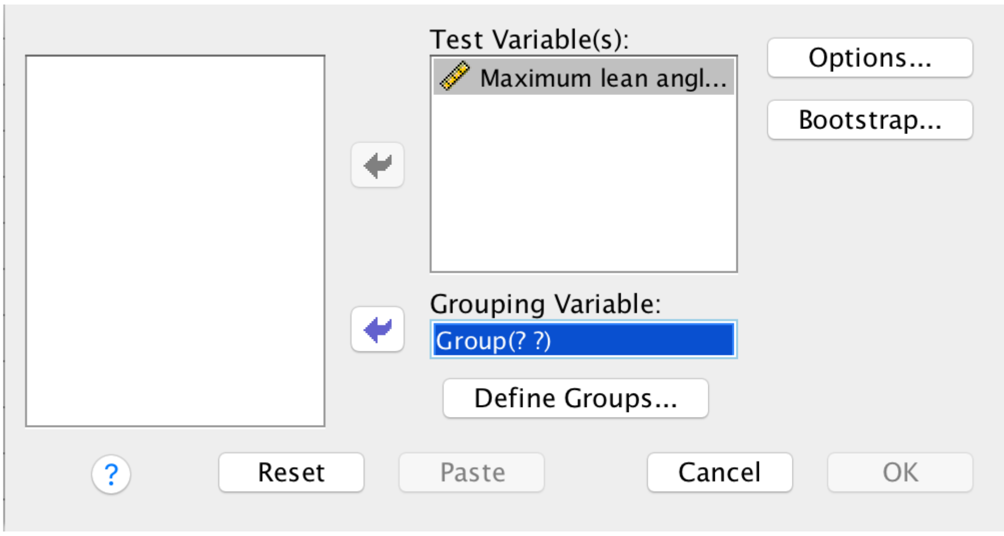


FIGURE 9.4: *Entering the variables.*

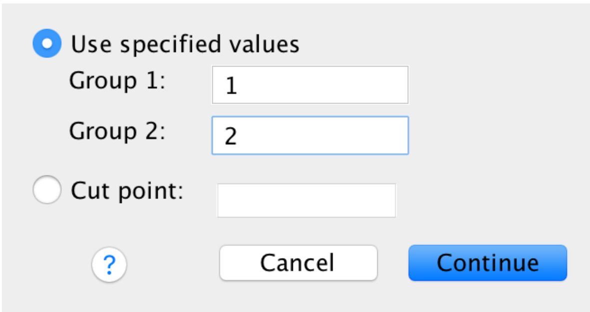


FIGURE 9.5: *Teling SPSS which groups to compare.*

9.4 Numerical summaries

The *mean*, *median*, *standard deviation* and *standard error* are produced as part of the output from the t-test analysis step above.

The *mean* and *standard error* for the *difference* is obtained from the t-test output too; see Sect. 9.3.

10

SPSS: graphs

In this chapter, you will learn to:

– Create graphs in SPSS.

The data files used in this section can be downloaded in Sect. 11.

You can download the data, and follow the instructions.

Graphs in SPSS are generally produced using the **Graphs** menu, and then selecting **Chart Builder...**

After making this selection, you usually get a warning (Fig. 10.1)). It is reminding you that it is **really important** that your have declared the **Measure** correctly for each variable in the **Variable View** (that is, correctly declared each variable as **Nominal**, **Ordinal** or **Scale**); see Sect. 7.

Provided you have done so, you can press **OK**.

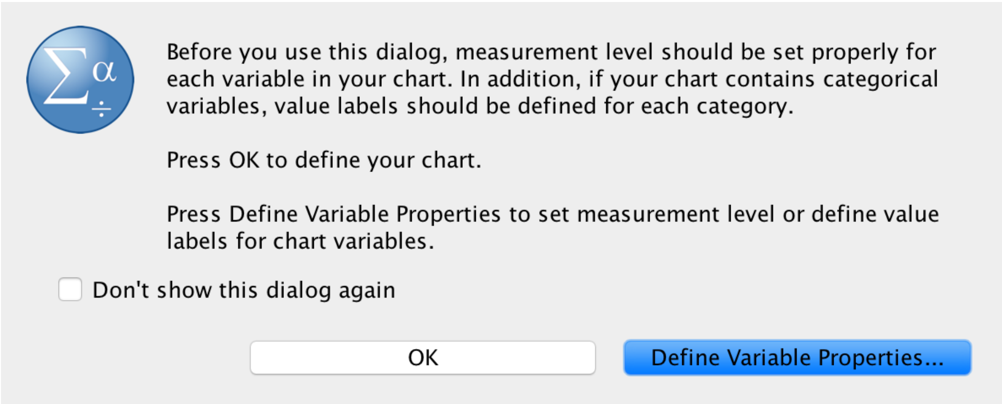


FIGURE 10.1: *The warning after selecting Graphs.*

The graphs produced by default in SPSS are usually an inconvenient size for copying-and-pasting into Word, PowewrPoint, and other places. We now explain the easy solution.

When graphs are created in SPSS using the **Chart Builder**, the window shown in Fig. 10.2 initially appears. Select the **Options** item from the right-hand side of this window.

Then, change the **Chart size** to (say) 60% or a similar value (Fig. 10.3). Now, when you copy-and-paste the graph, the sizing will be more appropriate.

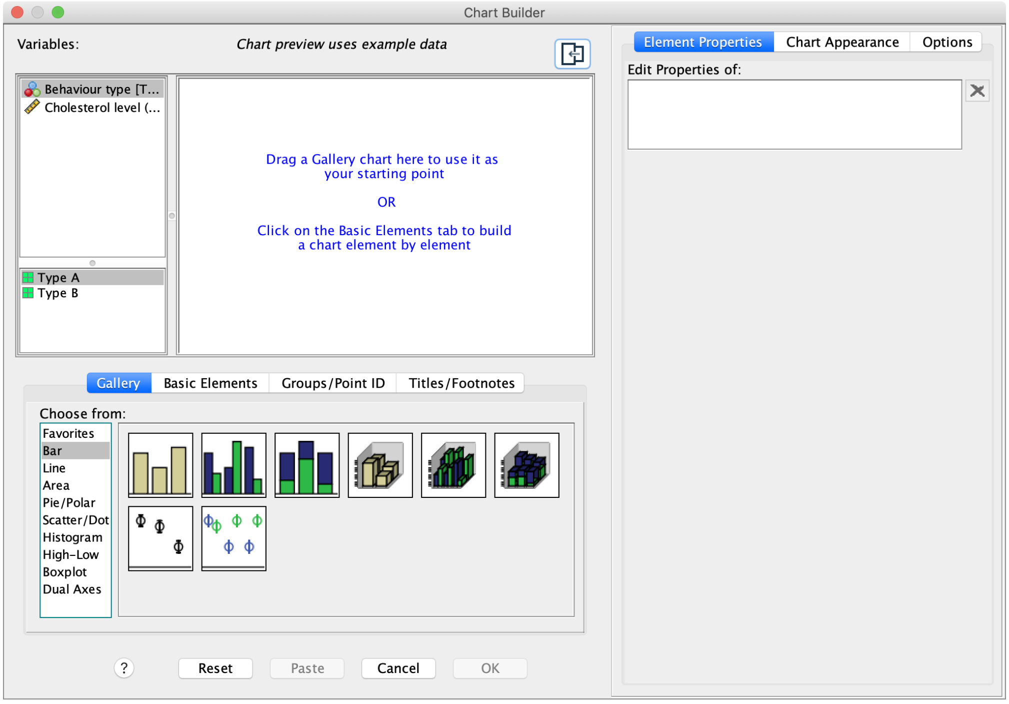


FIGURE 10.2: *The Chart Builder window.*

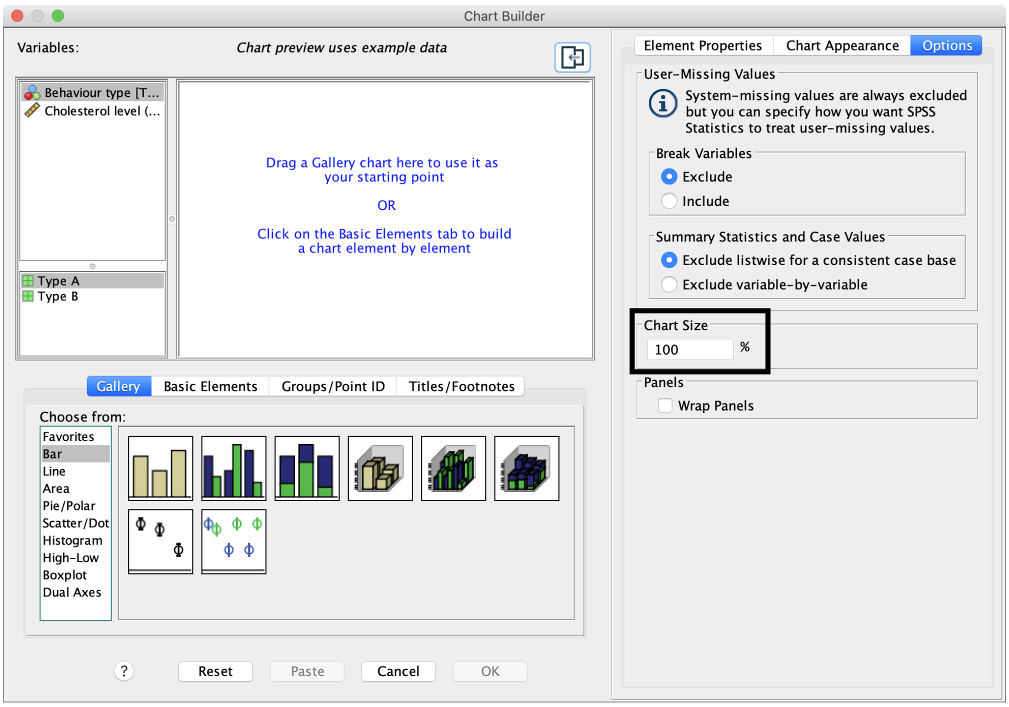


FIGURE 10.3: *The warning after selecting Graphs.*

10.1 Side-by-side bar charts

For this section, the data introduced in Sect. 8.1 are used.

1. **Use the SPSS menu:** select **Graphs> Chart Builder...**
2. **Select the type of plot**. Select **Bar** from the Gallery, then select the **second** template from the top row, and drag it to the canvas.

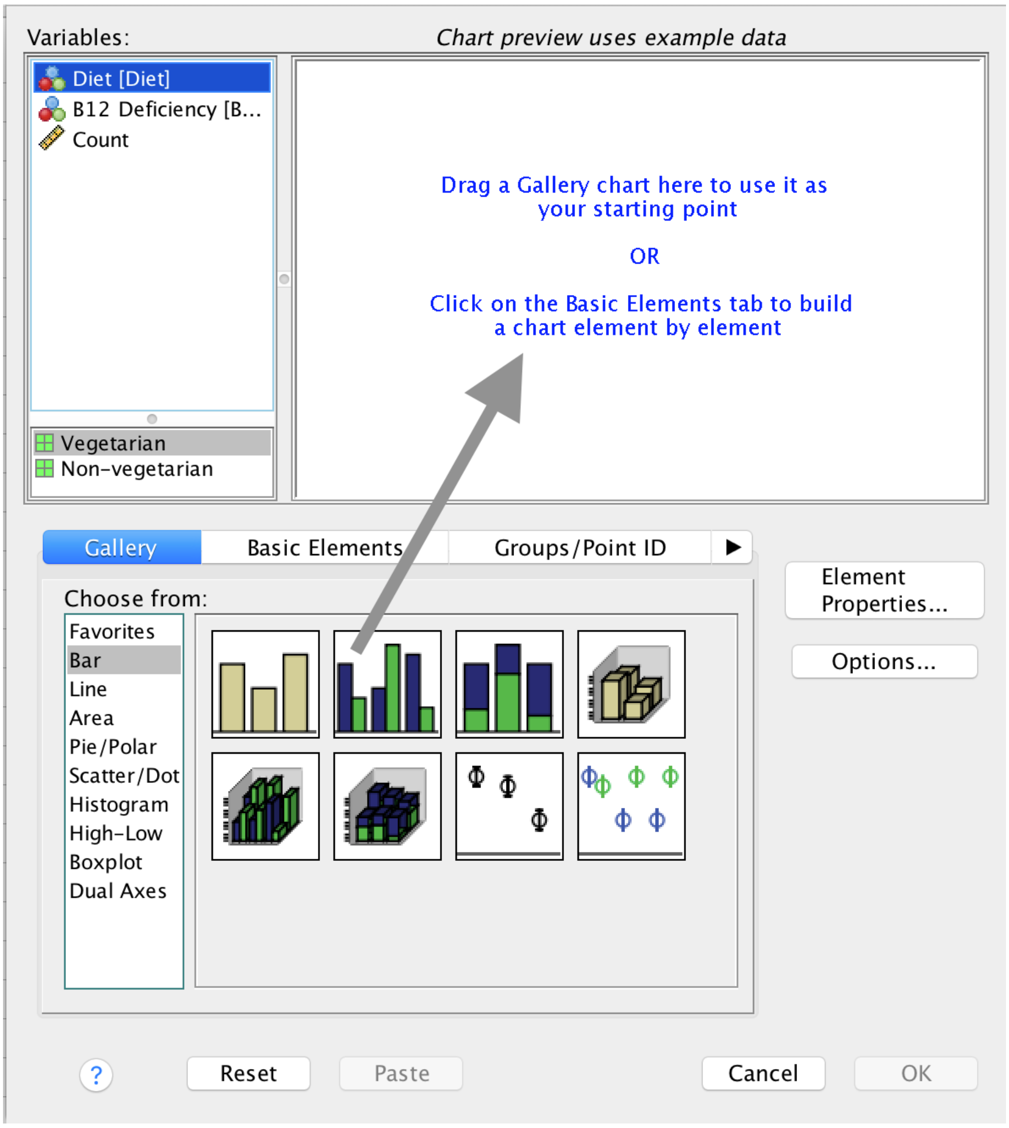


FIGURE 10.4: Selecting a side-by-side bar chart.

3. **Add the variables**. Drag one qualitative variable to the bottom (left-to-right) axis; this can often be seen as the explanatory variable. Drag the other to the top right (where it says, cryptically, **Cluster on X: set color**); see Fig. 10.5).
4. Press **OK**. You will have your graph in the SPSS Output window.

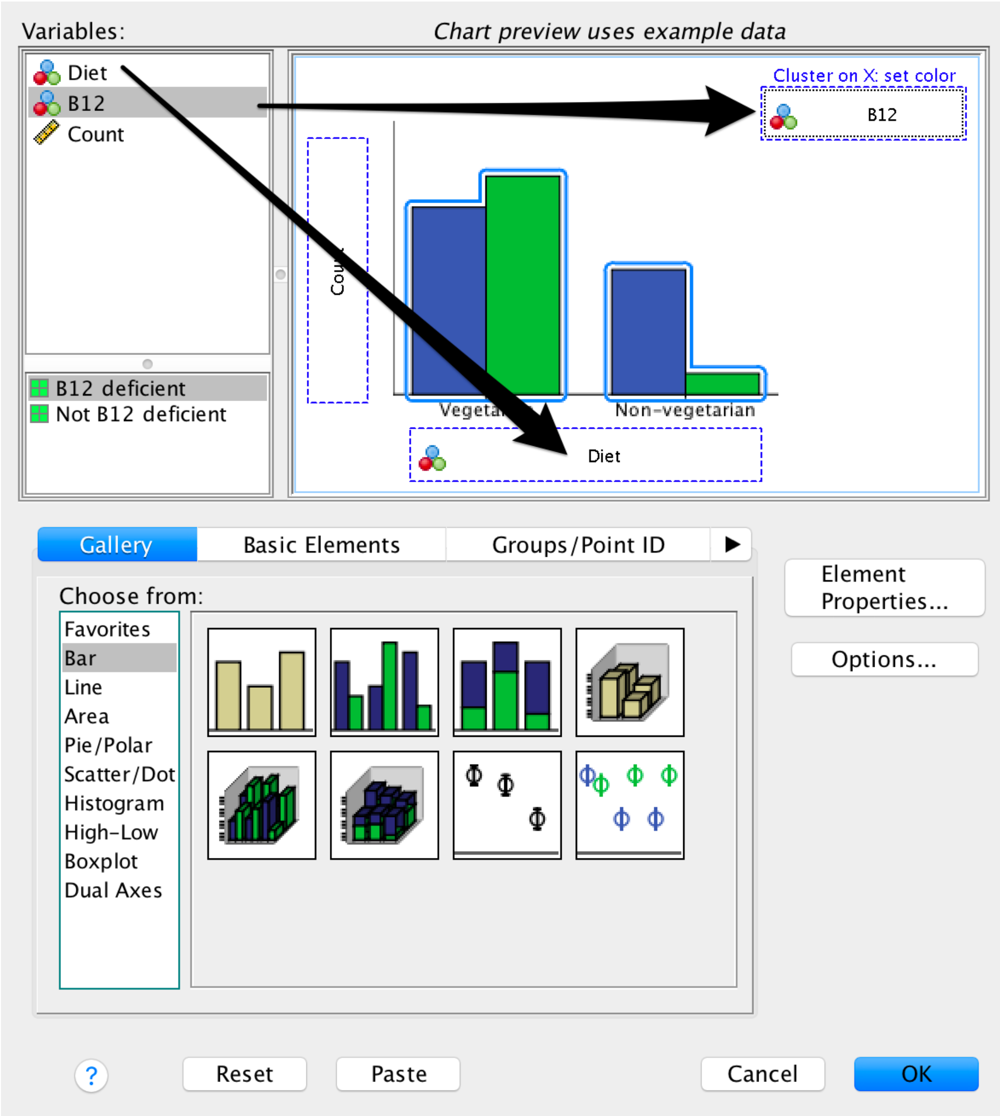


FIGURE 10.5: Adding the variables to create a side-by-side bar chart.

10.2 Stacked bar charts

For this section, the data introduced in Sect. 8.1 are used.

1. **Use the SPSS menu:** select **Graphs> Chart Builder…**
2. **Select the type of plot**. Select **Bar**, then select the **third** template from the top row, and drag it to the canvas (Fig. 10.6).
3. **Add the variables**. Drag one qualitative variable to the bottom (left-to-right) axis; this can often be seen as the explanatory variable. Drag the other to the top right (where it says, cryptically, **Stack: set color**); Fig. 10.7.
4. Press **OK**. You will have your graph in the SPSS Output window.

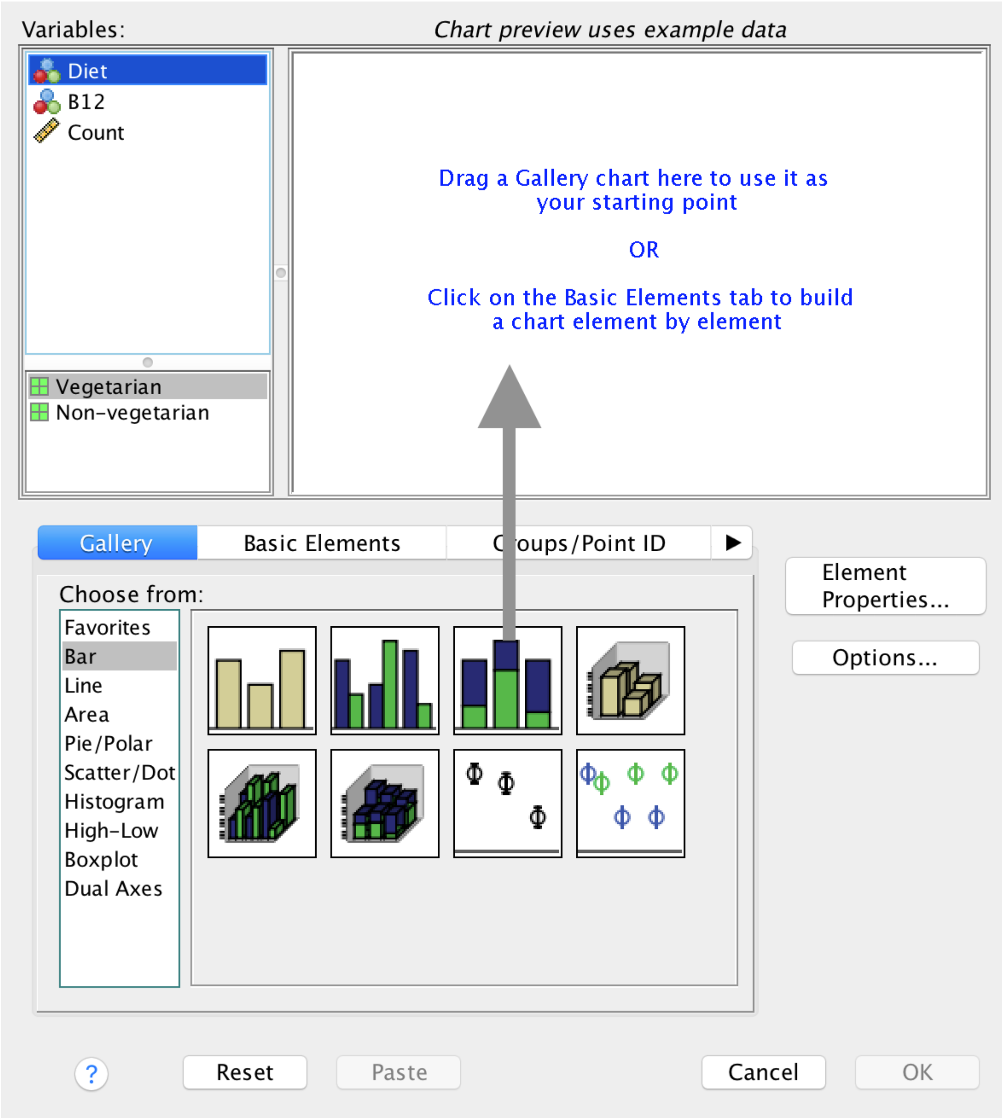


FIGURE 10.6: *Selecting a stacked bar chart.*

10.3 Boxplots

For this section, the data introduced in Sect. 9.1 are used.

1. **Use the SPSS menu:** select **Graphs> Chart Builder…**
2. **Select the type of plot**. Select **Boxplot**, then select the **first** template from the top row, and drag it to the canvas (Fig. 10.8).
3. **Add the variables**. Drag the qualitative variable to the bottom (left-to-right) axis; drag the quantitative variable to the vertical (up-and-down) axis (Fig. 10.9).
4. Press **OK**. You will have your graph in the SPSS Output window.

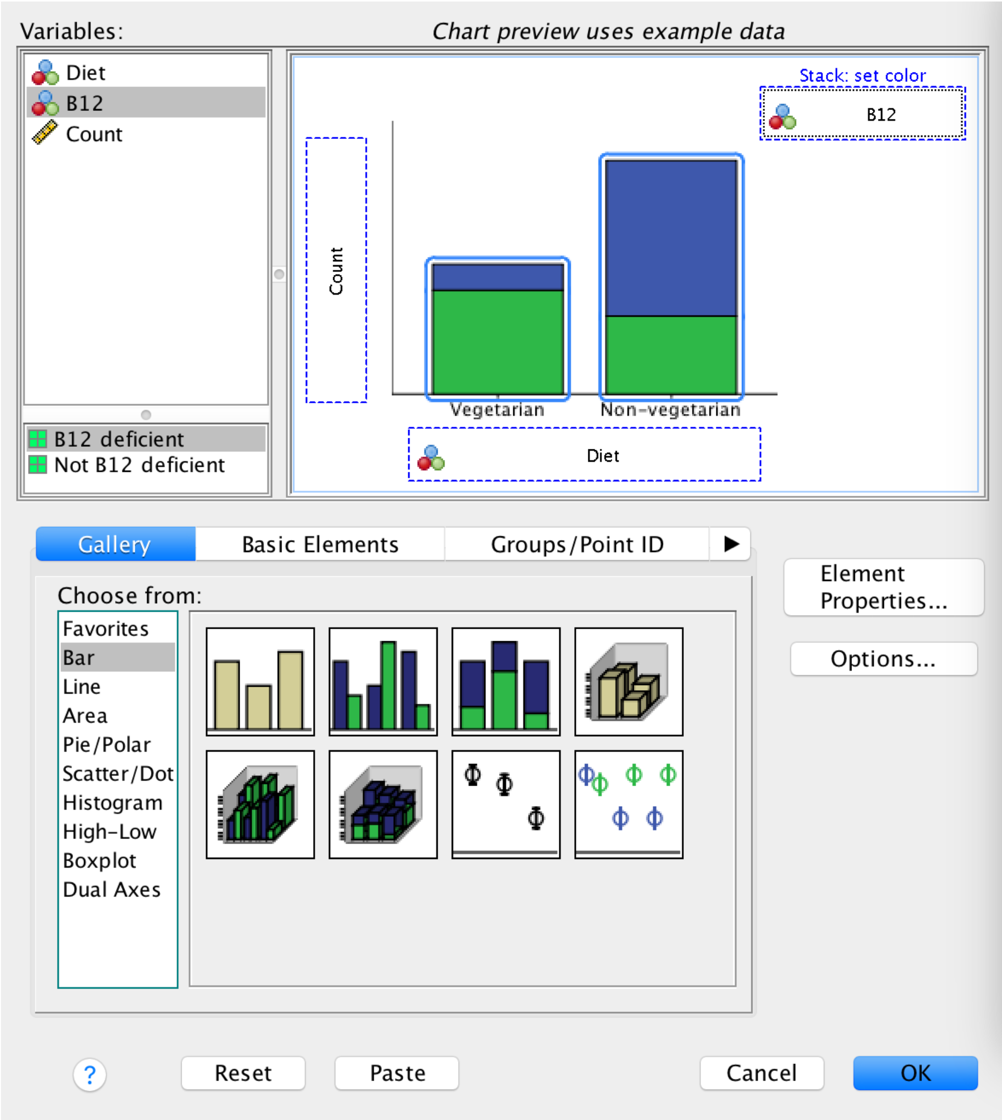


FIGURE 10.7: Adding the variables to create a stacked bar chart.

10.4 Error bar charts

For this section, the data introduced in Sect. 9.1 are used.

1. **Use the SPSS menu:** select **Graphs> Chart Builder...**
2. **Select the type of plot**. Select **Bar**, then select the **third** template from the **second row**, and drag it to the canvas (Fig. 10.10).
3. **Add the variables**. Drag the qualitative variable to the bottom (left-to-right) axis; drag the quantitative variable to the vertical (up-and-down) axis (Fig. 10.11).
4. Press **OK**. You will have your graph in the SPSS Output window.

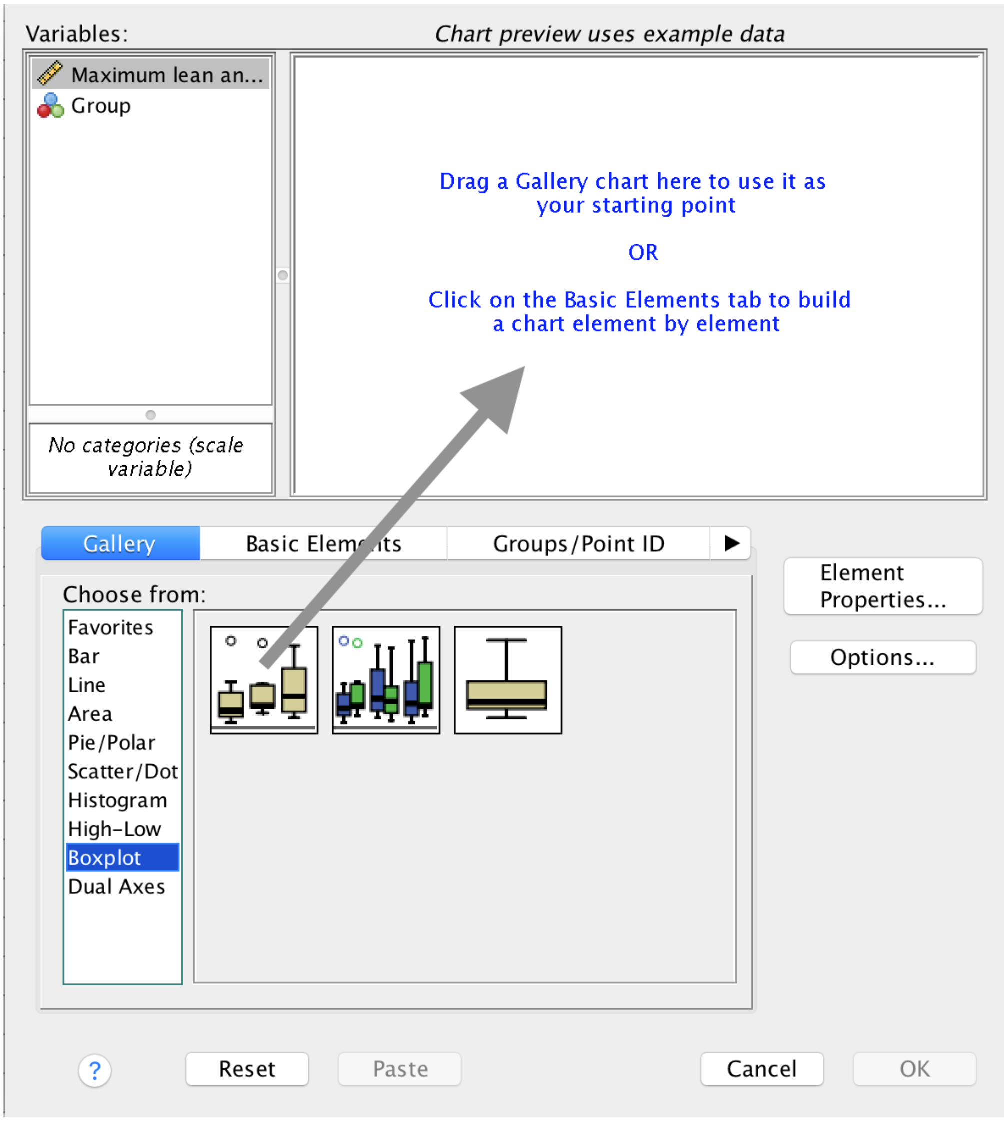


FIGURE 10.8: Selecting a boxplot.

10.5 Editing graphs

Your graph appears in the SPSS Output window. Double-clicking on the graph opens the **Chart Editor**.

In the **Chart Editor**, you can edit almost any part of the graph, by clicking on various elements of the graph (e.g. the axis labels) or clicking on the icons at the top of the Chart Editor window.

Once you have finished editing your graph in the **Chart Editor**, close that window, and the graph will be updated in the SPSS Output Window.

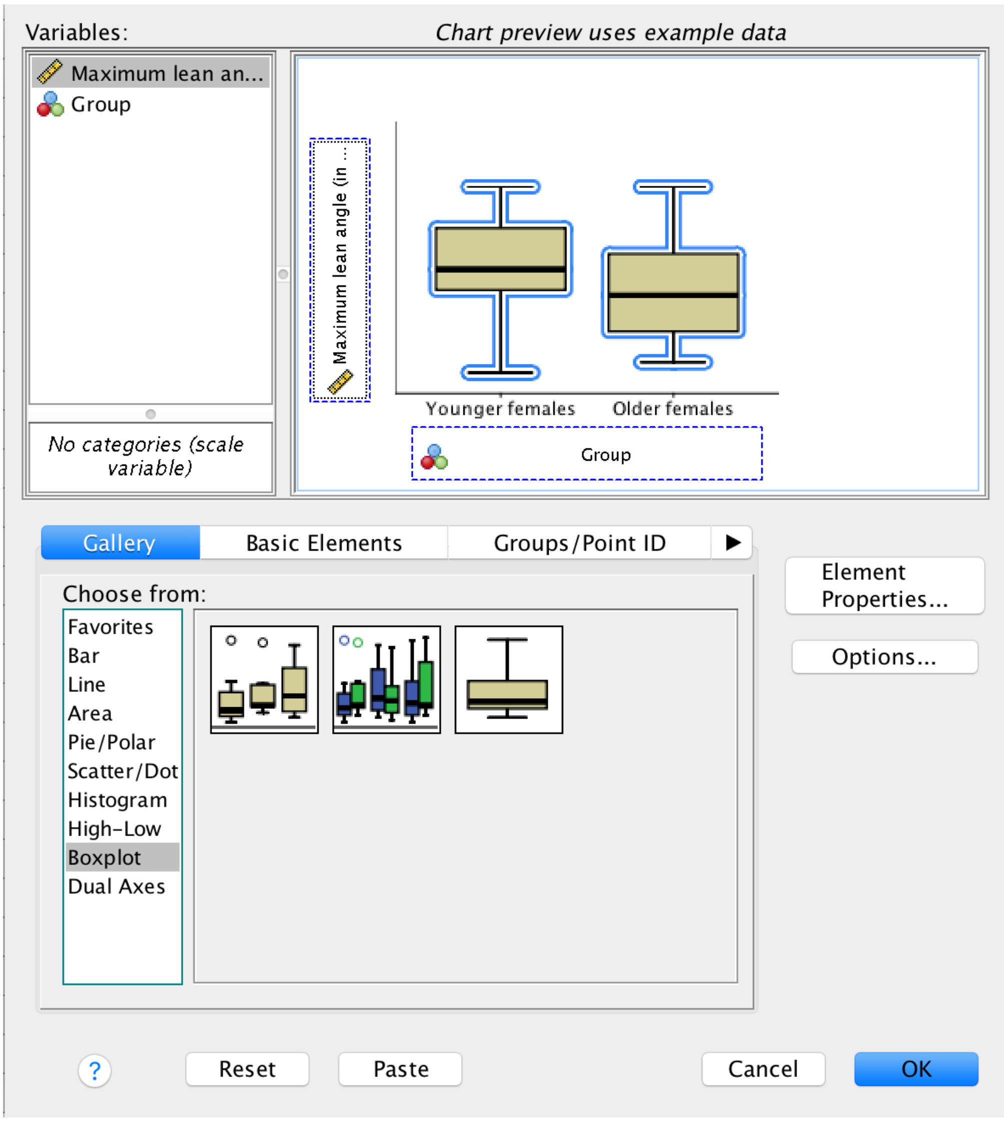


FIGURE 10.9: *Adding the variables to create a boxplot.*

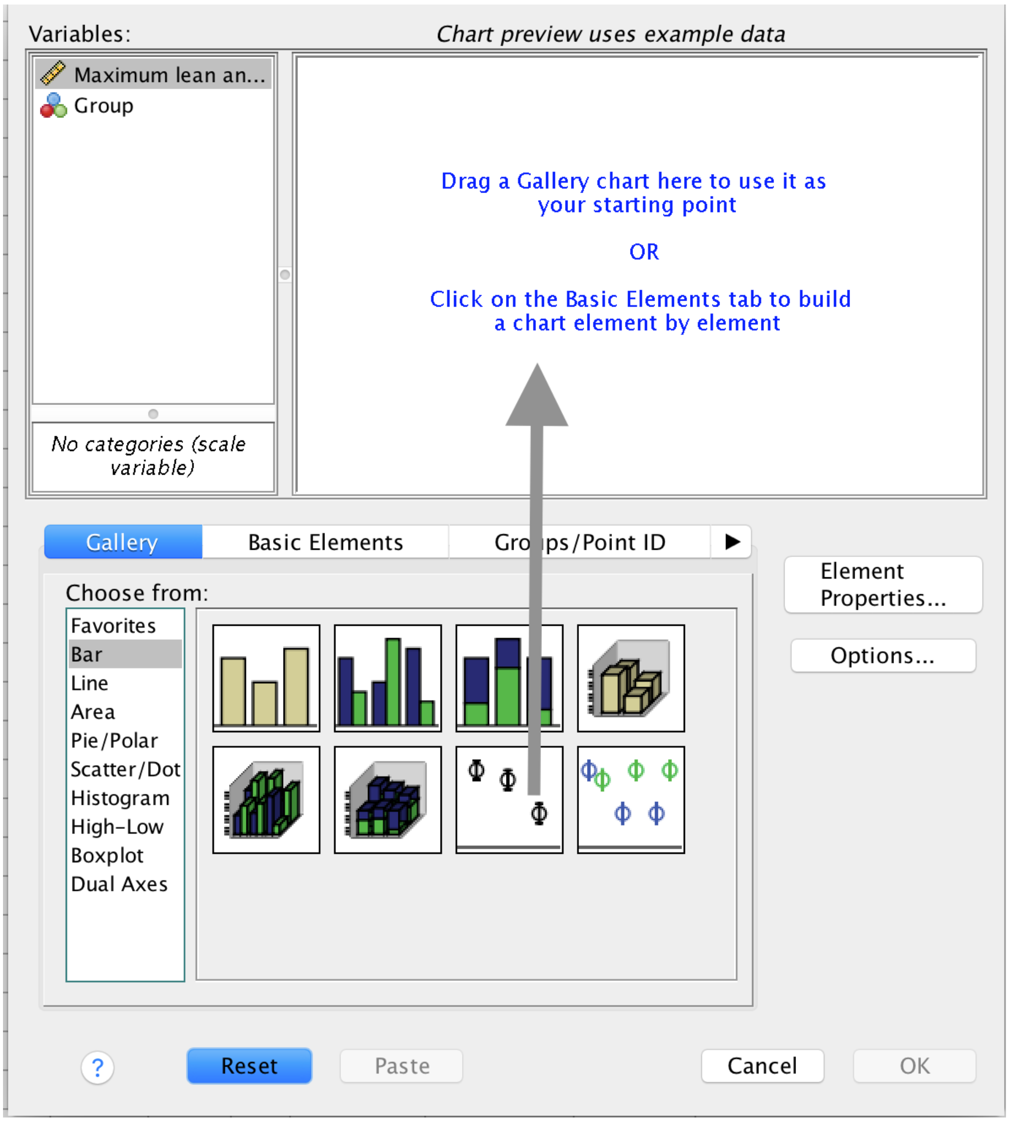


FIGURE 10.10: *Selecting an error bar chart.*

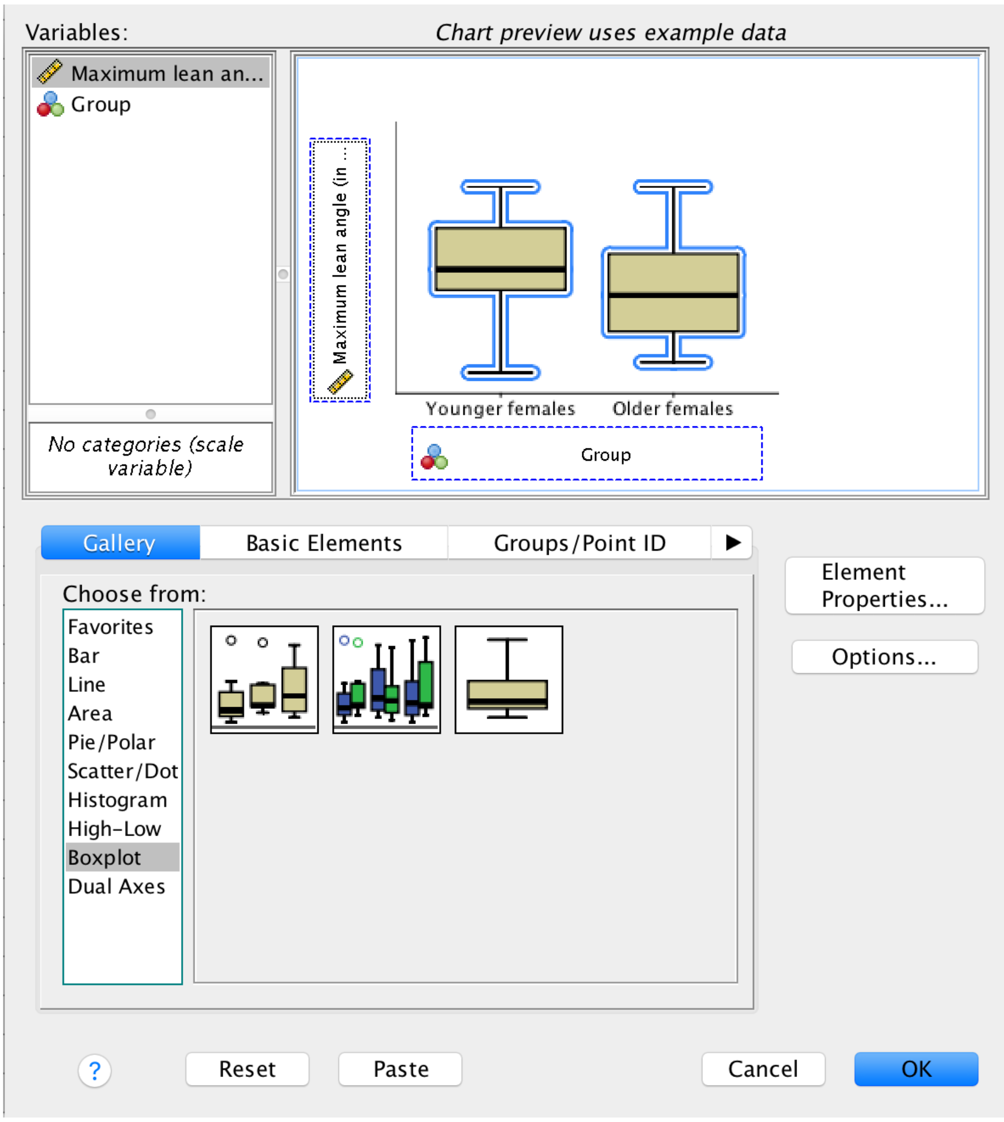


FIGURE 10.11: *Adding the variables to create an error bar chart.*

11

Data files

The following data sets are available for download.

Data	**Software**	**Reference**	**Download**
B12-Long data	jamovi	Sect. 5	`B12-Long.csv`[1]
B12-Long data	SPSS	Sect. 8	`B12-Long.sav`[2]
Face-plant data	jamovi	Sect. 6	`forwardfall.csv`[3]
Face-plant data	SPSS	Sect. 9	`forwardfall.sav`[4]

[1] ./Data/B12-Long.csv
[2] ./Data/B12-Long.sav
[3] ./Data/forwardfall.csv
[4] ./Data/forwardfall.sav

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